ET Pulse: AI-Synthesized Intelligence Briefing

ET Pulse is an advanced, AI-powered intelligence platform built to eliminate news noise and synthesize critical signals tailored exactly to the reader's persona. Built with Next.js and powered by Google's `gemini-2.5-flash`, the application tracks evolving story arcs and dynamically reconstructs complex economic news into actionable insights.

🚀 How It Works

ET Pulse operates as an intelligent middle-layer between raw news articles and the end user. Here is the operational workflow:

1. Data Aggregation & Stitching

The backend engine ingests multiple distinct articles covering the same macroeconomic event (e.g., a massive Union Budget announcement). Utilizing Google's Generative AI schema extraction, ET Pulse aggregates these sources, removes conflicting or redundant claims, and synthesizes one coherent "Story Arc."

2. The Interactive Timeline Engine

Once the events are stitched together, the frontend parses the data into an **Interactive Event Timeline**.

- The chronological sequence is color-coded using AI-derived **Sentiment Analysis** (Bullish, Bearish, or Neutral).
- Every event card acts as a drill-down dashboard that exposes a **Detailed Impact Analysis** explaining exactly *why* the update matters.
- Users can click **Trace Key Signal** to open the raw source article directly.

3. Graceful Rate-Limit Handling

To ensure a continuous user experience, the system's external requests are safely guarded by robust error boundaries. If the AI model hits an exhaustion or rate-limit threshold via the API, the backend seamlessly routes the user to an intelligent, rich fallback cache corresponding exactly to their active parameters.

🎯 Key Use Cases & Capabilities

The platform is designed to be highly dynamic, reacting to specific user constraints via prompt-engineering pipelines.

Persona-Driven Context

The exact same data payload is structurally rewritten depending on *who* is reading it:

- **Student Mode:** Explains complex financial jargon and highlights how macroeconomic shifts impact AI skilling, internship demand, and the future job market.
- **Investor Mode:** Eliminates narrative fluff and delivers a highly-dense, signal-heavy output focusing strictly on medium-term earnings, market rotation, risk visibility, and portfolio bias.
- **Founder Mode:** Reframes compliance adjustments and policy updates into actionable GTM (Go-To-Market) validation metrics and funding climates.

Native Vernacular Synthesis

Rather than performing a basic, literal dictionary translation, the **Vernacular Engine** prompts the Generative AI to apply *Cultural Adaptation*. For example, when switching the engine to **Telugu**, the application instantly re-stitches the briefing to use familiar analogies aimed at Indian audiences while keeping the tone informative and concise.

⚙️ Setup Instructions

To run **ET Pulse** locally for evaluation:

1. **Clone the repository:**

   ```
   git clone https://github.com/manidweep1306/ET-PULSE.git
   cd ET-PULSE/frontend
   ```

2. **Install Dependencies:** Ensure you have Node.js installed, then run:

   ```
   npm install
   ```

3. **Configure Environment Variables:** Create a new file named `.env.local` inside the `frontend/` directory and add your Gemini API Key safely:

   ```
   GEMINI_API_KEY=your_genai_api_key_here
   ```

4. **Launch the Application:** Start the Next.js development server:

   ```
   npm run dev
   ```

   Navigate to `http://localhost:3000` to interact with the platform safely.

🛠 Tech Stack

- **Frontend Layer:** Built using Next.js (React), powered by beautiful TailwindCSS designs (Glassmorphism, dark-modes, and CSS micro-animations).
- **AI Brain:** Hooked directly to `@google/generative-ai` leveraging structured JSON extraction via `gemini-2.5-flash`.